

Chow Siew Yeng

LANGUAGE ENGINEER · MACHINE LEARNING ENGINEER

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Experience

Temasek Lab@NTU

Singapore

RESEARCH ASSOCIATE

Sep. 2024 - present

- Designed and developed GenAI solution, including iterative workflow for continuous model finetuning to carry out qualitative data coding.
- Collaborated with domain experts (psychologists) to translate qualitative human codebooks into machine-executable prompt schemas.
- Explored various prompt-tuning methods such as self-improvement and incorporating API calls to larger models, and LLM fine-tuning with PEFT using multi-GPU clusters to increase F1 score of localisation task from 24% to 64% with less than 1000 data samples.
- Led the code base integration by reviewing, refactoring, and consolidating modules from 3 engineers into a unified, production-ready deliverable.
- Formulated evaluation framework for complex multi-label classification task to support performance monitoring
- Implemented RAG-based inferences in early versions of prototype.
- Built a scalable, multiprocessing Python data pipeline for high-throughput video preprocessing and feature extraction.
- Communicated system architectures, evaluation metrics, and technical progress with non-technical external stakeholders through monthly reviews.

AI Singapore

Singapore

AI APPRENTICE

Aug. 2023 - May 2024

- Developed LLM agents capable of conversing through Time Weighted Memory Retrieval and RAG to enable long-context conversational generations.
- Deployed agentic prototypes on Streamlit, enabling real-time user interactions while integrating external API calls with agentic workflow to produce natural language generations based on structured sources.
- Utilized prompt engineering techniques to steer LLMs towards generating targeted outputs, ensuring alignment with project goals.
- Fine-tuned an Inverse Text Normalization model using PyTorch Lightning and deployed a production-ready inference API via FastAPI, with progress tracking using Redis.
- Established MLOps and LLMOps standards using MLFlow for experiment tracking, Docker for containerization, and GitHub Actions for automated CI/CD testing.
- Authored detailed technical documentation of code in Read the Docs format, to facilitate knowledge sharing.

Transperfect

Singapore

LANGUAGE ENGINEER

Mar. 2022 - Aug. 2023

- Troubleshoot and resolved technical ASR system misrecognitions, engineering targeted data augmentations that improved end-to-end speech recognition accuracy and safety.
- Collaborated with global teams, doing code reviews on GitHub to improve digital assistant's proficiency in Singapore English and localised user experience.
- Acted as a subject matter expert, creating guidelines that drive locale-specific phonetic transcription for ASR, reducing speech recognition error rates in the region.
- Leveraged Docker for continuous testing, ensuring high quality of training dataset used for new model development.
- Assigned work to and guided a team of 3 data curators, fostering their skills in accurate transcription and data output.

A*STAR (Institute of Higher Performance Computing and Institute of Infocomm Research)

Singapore

INTERN

Jan. 2021 - Mar. 2021

- Trained CNN and MLP models for computer vision (CV) task of mapping facial landmarks to 7 distinct emotions, enhancing online learning tools in the COVID context.
- Researched on Discourse Representation Structure and evaluation benchmarks in Natural Language Inference (NLI), offering linguistic expertise to a query answering project.

Nanyang Technological University

Singapore

TEACHING ASSISTANT

Aug. 2020 - Dec. 2020

- Delivered comprehensive conceptual support in semantics and pragmatics to over 50 undergraduates, enhancing their academic understanding.
- Provided consultations and delivered weekly tutorials on concepts such as logical grammar, set theory and non-classical logics.
- Strategised lesson flow, guided discussion, graded group assignments and promoted understanding of linguistic concepts.

Nanyang Technological University

Singapore

RESEARCH ASSISTANT

Dec. 2017 - Mar. 2021

- Conducted in-depth analysis of over 3000 sentences, primarily focusing on grammatically incorrect sentences, to refine treebanking software for text parsing.
- Supervised adjudication between various pairs of annotators to facilitate inter-annotator agreement and quality of gold data.

Education

Nanyang Technological University

M.A. (RESEARCH) IN COMPUTATIONAL LINGUISTICS

- Built a rule-based syntactic parser for Singlish.

Singapore

Jan. 2020 - Dec. 2021

Nanyang Technological University

B.A. IN LINGUISTICS AND MULTILINGUAL STUDIES

- Graduated with Honours (Highest Distinction).
- Minor: Mathematics.

Singapore

Aug. 2015 - Jun. 2019

Skills

Machine Learning	LangChain, Transformers, PyTorch, Pandas, NumPy
NLP	NLTK, Regex
Deployment / DevOps	FastAPI, Docker, MLflow, GCP CLI
Programming	Python, React Native, LaTeX, Svelte
Languages	English, Cantonese, Mandarin, German (B2), Japanese (basic)

Publications

Eurali (LREC-COLING)

THIS WORD MEAN WHAT: CONSTRUCTING A SINGLISH DICTIONARY WITH CHATGPT

- Created new open-source resource for Singlish using ChatGPT through continuous API calls.
- Annotated and analysed findings according to results of annotation.
- Collaborated on dictionary front-end using typescript and svelte.

Torino, Italy

May 2024

LREC

SINGLISH WHERE GOT RULES ONE

- Developed a formal grammar of Singlish that can generate syntactic trees and DMRS from input sentences.
- Increased accuracy in parsing Singlish sentences from 46.7% to 70.0%.
- Presented findings and international DELPH-IN summits and LREC 2022.

Marseille, France

May 2022

Additional Courses

2023	German B2 , Goethe Institute
2022	Mathematics for Machine Learning Specialization , Coursera
2022	Natural Language Processing with Probabilistic Models , DeepLearning.AI
2022	Natural Language Processing with Classification and Vector Spaces , DeepLearning.AI
2021	German Summer Course by Technische Universität Dresden , scholarship awarded by DAAD
2021	11th Lisbon Machine Learning School , Instituto Superior Técnico